Cybersecurity Investigator

Cadette Cybersecurity Badge — Girls Build Bots

Presented by your robotics mentors

The capstone

Today you run a **full incident investigation** — thinking like both:

- The **attacker** — how was this done?
- The **defender** — how do we prove it, and prevent it?

The discipline: no conclusion outruns its support.

How real investigations work

1. **Secure** the evidence
2. Build a **timeline**
3. Form **hypotheses**
4. **Test** each against the evidence
5. *Only then* conclude

Two rules for today 📜

1. Every claim needs an exhibit.

2. The goal is the attack vector — how it happened — not just a culprit.

And fair warning: this file contains a red herring. Real cases always do. 🙂

The Case 📁

“The Bake Sale Account Takeover”

A club’s social account posted spam overnight.

Your team gets the full file. Work it.

Step 1 — Log every exhibit 

Exhibit	What it shows	Corroborated by	Confidence
			high / med / low

Confidence matters as much as the finding.

Step 2 — Build the timeline 🕐

Lay the strips out physically:

DM → click → fake login → 2:14 AM access → spam posts

Step 3 — Form hypotheses 🔬

At least two — including the suspicious rival club member.

For each one: - Which exhibits **support** it? - Which exhibits **contradict** it?

Step 4 — Name the attack vector 🎯

phishing link → credential theft → account takeover
…with the **evidence chain** that proves it.

When the red herring collapses 

Someone “seemed suspicious” — but the evidence doesn’t hold.

Suspicion isn’t proof.

That collapse is the most important moment of the session.

The Incident Report 

Like professionals:

1. **What happened** — the timeline
2. **How** — attack vector + evidence chain
3. **What we can't prove** — state unknowns honestly. Real reports always have them.
4. **Remediation** — three defenses that would have stopped this

Remediation ties the series together 🔗

- **Basics** explained the system
- **Safeguards** taught the defenses
- **Investigator** shows exactly where they'd have worked

2FA · a URL-checking habit · verify before clicking

Case Presentations ⚖️

Five minutes per team. Then **one cross-examination question**:

"How do you know the login wasn't the treasurer traveling?"

Answer from exhibits.

Robot moment 

We’ve debugged a mystery robot failure exactly this way — logs, timeline, hypotheses tested one at a time.

And the first suspect is often innocent.

Pick your habit 

Which investigative habit are you keeping?

- Evidence before conclusions
- Corroborate everything
- Beware the obvious suspect

What you accomplished ✅

- Built an evidence log and timeline from an incident file
- Tested multiple hypotheses and rejected one that failed
- Identified an attack vector and supported it with an evidence chain
- Wrote remediation recommendations and defended them under questioning

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